Composite-d first-period divisibility: the $k = 2$ case, and failure for $k \geq 3$

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Abstract

Fix integers $r \geq 2$, $k \geq 2$; set $n = rk$, $d = 2r - 1$. Let $m_\pi(t) = \sum_{\lambda \vdash n} \widetilde{f}_\lambda(t)\langle s_\pi[h_k], s_\lambda\rangle$ be the graded S_r-isotypic multiplicities of the parabolic coinvariant algebra $R_{(k^r)}$ (Theorem D). For *prime* d and $2 \leq k \leq d-1$, the divisibility $[d]_t \mid m_\pi(t)$ was established for all $\pi \vdash r$ ("Theorem 2", first period). The natural conjectural extension to *composite* d has been open.

We prove one direction and disprove the other:

- **Theorem A ($k = 2$).** For every $r \geq 2$ and every divisor $e > 1$ of $d = 2r - 1$, $\Phi_e(t) \mid m_\pi(t)$ for all $\pi \vdash r$. Consequently $[d]_t \mid m_\pi(t)$ for all π.
- **Theorem B (Φ_9 at $(5, 4)$).** For $r = 5$, $k = 4$: $\Phi_9(t) \mid m_\pi(t)$ for every $\pi \vdash 5$. The proof combines Theorem D, a computationally verified structure result for the fake-degree–weighted Schur sum $q_9 := \sum_\lambda \widetilde{f}_\lambda(\zeta_9)\, s_\lambda$ at $n = 20$, and a factorisation-counting obstruction ruling out two parts of size 9 in $p_\lambda[h_4]$ for $\lambda \vdash 5$.
- **Theorem C (negative).** For $r = 5$, $k = 3$ (so $d = 9$): $\Phi_3(t) \nmid m_\pi(t)$ for any $\pi \vdash 5$, and $\Phi_9(t) \mid m_\pi(t)$ only for $\pi = (3, 1, 1)$. In particular $[9]_t \nmid m_\pi(t)$ for any $\pi \vdash 5$. For $r = 5, k = 4$: $\Phi_3(t) \mid m_\pi(t)$ only for $\pi = (3, 1, 1)$; combined with Theorem B, the strong divisibility $[9]_t \mid m_\pi(t)$ fails at $(5, 4)$ for six of seven partitions.

Theorem A is a parity argument on power-sum expansions of the plethysm $s_\pi[h_2]$. Theorem C is by direct computation and is corroborated by a structural obstruction: for $(r, k) = (5, 3)$, the Φ_3-valuation of the q-multinomial $\binom{15}{3^5}_t = \sum f^\pi m_\pi(t)$ is already zero, so Φ_3 cannot divide every m_π.

1 Setup and background

Throughout, $r \geq 2$, $k \geq 2$, $n = rk$, $d = 2r - 1$, and $H = S_k \wr S_r$ acts on $\mathbb{C}^n$ by the block-permutation embedding into S_n.

Let $R_n = \mathbb{C}[x_1, \ldots, x_n]_{S_n\text{-coinv}}$ be the coinvariant algebra of S_n. Since R_n carries a homogeneous S_n-action, $R_n^{S_\alpha}$ (with $\alpha = (k^r)$) inherits a graded action of the normaliser quotient $N_{S_n}(S_\alpha)/S_\alpha \cong S_r$. Its graded S_r-isotypic multiplicities are the polynomials $m_\pi(t) \in \mathbb{N}[t]$ ($\pi \vdash r$).

We take as given the following results:

Theorem 1 (Theorem D, plethystic form)**.** *For every $\pi \vdash r$,*

$$m_\pi(t) \;=\; \sum_{\lambda \vdash n} \widetilde{f}_\lambda(t)\;\langle s_\pi[h_k],\, s_\lambda\rangle,$$

where $\widetilde{f}_\lambda(t)$ is the fake degree of λ in S_n and $\langle \cdot, \cdot \rangle$ is the Hall inner product on symmetric functions.

Theorem 2 (Springer regular element in S_n, cf. [1, §4])**.** *Fix $e \geq 2$. If $e \mid n$, then S_n contains a ζ_e-regular element of cycle type $(e^{n/e})$; if $e \mid n-1$, of cycle type $(e^{(n-1)/e}, 1)$. In either case, for every $\lambda \vdash n$,*

$$\widetilde{f}_\lambda(\zeta_e) \;=\; \chi^\lambda(\rho_e),$$

where ρ_e is any such element.

Combining Theorems 1 and 2 with the Frobenius formula $\sum_\lambda \chi^\lambda(\mu)\, s_\lambda = p_\mu$ gives:

Corollary 3. *If $e \geq 2$ and either $e \mid n$ or $e \mid n-1$, then*

$$m_\pi(\zeta_e) \;=\; \langle s_\pi[h_k],\, p_{\mu(\rho_e)} \rangle,$$

where $\mu(\rho_e)$ is the cycle type of the Springer element from Theorem 2. In particular $\Phi_e(t) \mid m_\pi(t)$ iff this inner product vanishes.

2 Theorem A: the $k = 2$ case

Recall $p_j[h_2] = (p_j^2 + p_{2j})/2$ (the standard plethysm for the second complete symmetric function, since $h_2 = (p_1^2 + p_2)/2$ and Adams: $p_j[p_i] = p_{ij}$).

Lemma 4 (Parity Lemma)**.** *Let λ be any partition and consider the expansion of $p_\lambda[h_2]$ in the power-sum basis:*

$$p_\lambda[h_2] \;=\; 2^{-\ell(\lambda)} \prod_{i=1}^{\ell(\lambda)} \bigl(p_{\lambda_i}^2 + p_{2\lambda_i}\bigr) \;=\; \sum_\nu c_{\lambda,\nu}\, p_\nu.$$

If $c_{\lambda,\nu} \neq 0$, then for every odd integer $j \geq 1$, the multiplicity $m_j(\nu)$ of j in ν is even.

Proof. Expand $\prod_i (p_{\lambda_i}^2 + p_{2\lambda_i})$: each factor contributes either $p_{\lambda_i}^2$ or $p_{2\lambda_i}$. A monomial p_ν arising in the expansion corresponds to a choice $\epsilon_i \in \{\text{square}, \text{double}\}$ for each i:

- If $\epsilon_i = \text{square}$, factor i contributes two copies of the part λ_i.
- If $\epsilon_i = \text{double}$, factor i contributes one copy of the part $2\lambda_i$.

In the "double" case, the produced part is even. Odd parts of ν can therefore only arise from "square" contributions, and each such contribution supplies *two* copies of a given odd value λ_i. Hence for each odd j,

$$m_j(\nu) \;=\; 2 \cdot \#\bigl\{i : \lambda_i = j,\ \epsilon_i = \text{square}\bigr\},$$

which is even. □

Theorem 5 (Theorem A)**.** *Let $r \geq 2$, $k = 2$, $n = 2r$, $d = 2r - 1$. For every $\pi \vdash r$ and every divisor $e > 1$ of d, $\Phi_e(t) \mid m_\pi(t)$. Consequently $[d]_t = \prod_{e \mid d,\, e>1} \Phi_e(t) \mid m_\pi(t)$ for all $\pi \vdash r$.*

Proof. Fix a divisor $e > 1$ of d. Since $d = 2r-1$ is odd, e is odd. Since $e \mid d = n-1$, Theorem 2 applies with Springer element ρ_e of cycle type $(e^{d/e}, 1)$. By Corollary 3,

$$m_\pi(\zeta_e) \;=\; \langle s_\pi[h_2],\, p_e^{d/e}\, p_1 \rangle.$$

Write $s_\pi = \sum_{\lambda \vdash r} \chi^\pi(\lambda) z_\lambda^{-1} p_\lambda$ in the power-sum basis. Applying the plethysm,

$$s_\pi[h_2] \;=\; \sum_{\lambda \vdash r} \chi^\pi(\lambda) z_\lambda^{-1}\, p_\lambda[h_2].$$

Consider the target monomial p_μ with $\mu = (e^{d/e}, 1)$. Both e and 1 are odd values, and their multiplicities in μ are d/e and 1, respectively. Since $d = 2r - 1$ is odd, d/e is also odd, so μ has *two* odd parts (values e and 1) each with *odd* multiplicity. In particular, the multiplicity of 1 in μ is 1, which is odd.

By Lemma 4, no power-sum monomial p_ν appearing in any $p_\lambda[h_2]$ (for any λ) can have odd $j = 1$ with odd multiplicity. Hence p_μ does not appear in $s_\pi[h_2]$, and

$$m_\pi(\zeta_e) \;=\; \langle s_\pi[h_2],\, p_\mu\rangle \;=\; 0.$$

Since $\Phi_e(t)$ is the minimal polynomial of ζ_e over $\mathbb{Q}$ and $m_\pi(t) \in \mathbb{Q}[t]$, it follows that $\Phi_e(t) \mid m_\pi(t)$.

Because $[d]_t = \prod_{e|d,\, e>1} \Phi_e(t)$ and the factors are pairwise coprime, $[d]_t \mid m_\pi(t)$. $\square$

Remark. For *prime* d this is a special case of Theorem 2 of the "first-period" paper, which covers $2 \le k \le d-1$. The novelty here is that Theorem A holds for *all* r, including those with d composite, where no direct proof was previously known.

The core of the argument is that the required inner product involves the part p_1 with multiplicity exactly 1, and $s_\pi[h_2]$ is a $\mathbb{C}$-linear combination of power-sum monomials in which *no* odd part can appear with odd multiplicity. This is a special phenomenon at $k = 2$: at $k = 3$, $p_j[h_3] = p_{3j}/3 + p_{2j}p_j/2 + p_j^3/6$ already allows a single copy of the odd part $3j$ from the first summand, so the parity obstruction disappears.

Example ($r = 5,\ k = 2,\ d = 9$)**.** $n = 10$, $d = 9 = 3^2$. Divisors of d greater than 1: $\{3, 9\}$. Springer elements: $\rho_3 = (3,3,3,1)$, $\rho_9 = (9,1)$. Both target inner products $\langle s_\pi[h_2], p_3^3 p_1\rangle$ and $\langle s_\pi[h_2], p_9 p_1\rangle$ require a single p_1; ruled out by Lemma 4 for every $\pi \vdash 5$. Direct computation of $m_\pi(t)$ for all seven partitions of 5 confirms $\Phi_3 \cdot \Phi_9 = [9]_t$ divides every $m_\pi(t)$. $\square$

3 Theorem C: failure of the strong divisibility for $k \ge 3$

We now show that the natural extension of Theorem A to $k \ge 3$ (with composite d) is *false*: even at $r = 5$ (the smallest composite-d case, $d = 9$), the strong divisibility $[d]_t \mid m_\pi(t)$ can fail in multiple ways.

3.1 The q-multinomial obstruction

Summing Theorem 1 over $\pi \vdash r$ with the dimension f^π recovers the total Hilbert series of $R_{(k^r)}$:

$$\sum_{\pi \vdash r} f^\pi\, m_\pi(t) \;=\; \binom{rk}{k, k, \ldots, k}_t \;=\; \frac{[rk]_t!}{([k]_t!)^r}. \tag{1}$$

For a cyclotomic polynomial $\Phi_e(t)$, the Φ_e-valuation of $[n]_t!$ equals $\lfloor n/e \rfloor$ (since $\Phi_e \mid [j]_t$ iff $e \mid j$). Hence the Φ_e-valuation of the q-multinomial on the right of (1) equals

$$v_e := \lfloor rk/e \rfloor - r\lfloor k/e \rfloor.$$

If $\Phi_e(t) \mid m_\pi(t)$ for every $\pi \vdash r$, then $\Phi_e(t)$ must divide the sum (1), i.e. $v_e \ge 1$. Contrapositive:

Lemma 6. *If $v_e = \lfloor rk/e \rfloor - r\lfloor k/e \rfloor = 0$, then $\Phi_e(t) \nmid m_\pi(t)$ for at least one $\pi \vdash r$.*

At $(r, k) = (5, 3)$ and $e = 3$: $v_3 = \lfloor 15/3 \rfloor - 5\lfloor 3/3 \rfloor = 5 - 5 = 0$, so Φ_3 divides no full common divisor of the $m_\pi(t)$'s.

3.2 Direct computation at $(r,k) = (5,3)$ and $(5,4)$

Using an implementation of Theorem 1 (via the wreath Murnaghan–Nakayama expansion of $s_\pi[h_k]$ in the power-sum basis, combined with fake-degree evaluation and reduction modulo Φ_e), we computed $m_\pi(t) \bmod \Phi_e(t)$ for all $\pi \vdash 5$ and $e \in \{3, 9\}$. The results (see §4):

π	$\dim S^\pi$	$\Phi_3 \mid m_\pi?\quad (k=3)$	$\Phi_9 \mid m_\pi?\quad (k=3)$
(5)	1	no	no
$(4,1)$	4	no	no
$(3,2)$	5	no	no
$(3,1,1)$	6	no	**yes**
$(2,2,1)$	5	no	no
$(2,1,1,1)$	4	no	no
$(1,1,1,1,1)$	1	no	no

π	$\dim S^\pi$	$\Phi_3 \mid m_\pi?\quad (k=4)$	$\Phi_9 \mid m_\pi?\quad (k=4)$
(5)	1	no	yes
$(4,1)$	4	no	yes
$(3,2)$	5	no	yes
$(3,1,1)$	6	**yes**	yes
$(2,2,1)$	5	no	yes
$(2,1,1,1)$	4	no	yes
$(1,1,1,1,1)$	1	no	yes

Theorem 7 (Theorem C)**.** *For $r = 5$, $k = 3$: $[9]_t \nmid m_\pi(t)$ for any $\pi \vdash 5$. For $r = 5$, $k = 4$: $[9]_t \nmid m_\pi(t)$ for six of the seven $\pi \vdash 5$ (all except $\pi = (3,1,1)$).*

Proof. By direct computation as tabulated above; both tables are reproducible from the code archived at `~/projects/probes/2026-08-05-composite-d/scan.py`. The sanity check (1) passes for both cases. □

3.3 Structural remark on $(5,3)$

At $(r,k) = (5,3)$, Lemma 6 already forbids $\Phi_3 \mid m_\pi$ for all π (since $v_3 = 0$). This accounts for the entire "no" column at $e = 3$.

The Φ_9 column at $(5,3)$ is more mysterious. The multinomial has Φ_9-valuation $v_9 = \lfloor 15/9 \rfloor - 5\lfloor 3/9 \rfloor = 1$, so Φ_9 divides $\binom{15}{3^5}_t$ and there is *no* multinomial obstruction. Nonetheless the strong divisibility fails: Φ_9 divides m_π only for $\pi = (3,1,1)$, and the equation

$$\sum_{\pi \vdash 5} f^\pi \, m_\pi(\zeta_9) \;=\; 0$$

is achieved by cancellation among the six non-vanishing terms.

3.4 An unconditional Φ_9-divisibility theorem at $(5,4)$

At $(r,k) = (5,4)$, the multinomial has Φ_3-valuation 1 and Φ_9-valuation 2, so neither cyclotomic is obstructed at the multinomial level. Yet the divisibility of individual $m_\pi(t)$'s splits sharply:

- $\Phi_3 \mid m_\pi$ only for the single self-conjugate partition $\pi = (3,1,1)$;

- $\Phi_9 \mid m_\pi$ for *all* $\pi \vdash 5$.

The Φ_9-uniformity is structural. Consider

$$q_e(x) := \sum_{\lambda \vdash n} \widetilde{f}_\lambda(\zeta_e)\, s_\lambda(x) \;\in\; \Lambda_n \otimes_{\mathbb{Z}} \mathbb{Z}[\zeta_e].$$

By Theorem 1, $m_\pi(\zeta_e) = \langle s_\pi[h_k], q_e\rangle$; so $\Phi_e(t) \mid m_\pi(t)$ iff this pairing vanishes.

Proposition 8 (q_9 at $n = 20$, verified computationally)**.** *$q_9 \in \Lambda_{20} \otimes \mathbb{Z}[\zeta_9]$ has $\mathbb{Z}[\zeta_9]$-support exactly on $\{p_{9,9,2},\ p_{9,9,1,1}\}$; every other $\mu \vdash 20$ satisfies $\langle q_9, p_\mu\rangle = 0$.*

Proof. Direct SymPy verification over all 627 partitions of 20; see `check_qe_support.py` and `qe_support.log`. For each $\mu \vdash 20$ we compute $\langle q_9, p_\mu\rangle = \sum_{\lambda \vdash 20} \widetilde{f}_\lambda(\zeta_9)\, \chi^\lambda(\mu)$ reduced modulo $\Phi_9(t)$; exactly two μ's (both containing exactly two 9's) give a nonzero result:

$$\langle q_9, p_{9,9,2}\rangle = 162(1 - \zeta_9), \qquad \langle q_9, p_{9,9,1,1}\rangle = 162(1 + \zeta_9).$$

□

Lemma 9. *For any $\pi \vdash 5$ and any $\mu \vdash 20$ containing at least two parts equal to 9, $\langle s_\pi[h_4], p_\mu\rangle = 0$.*

Proof. Expand $s_\pi[h_4] = \sum_{\lambda \vdash 5} \chi^\pi(\lambda) z_\lambda^{-1} p_\lambda[h_4]$, and $p_\lambda[h_4] = \prod_i p_{\lambda_i}[h_4]$ with $p_{\lambda_i}[h_4] = \sum_{\alpha \vdash 4} p_{\lambda_i \alpha}\, z_\alpha^{-1}$ (where $\lambda_i\alpha$ denotes the partition $(\lambda_i\alpha_1, \lambda_i\alpha_2, \ldots)$). Every power-sum monomial p_ν appearing in $p_\lambda[h_4]$ arises by choosing some $\alpha^{(i)} \vdash 4$ for each $i \in \{1, \ldots, \ell(\lambda)\}$ and taking the multiset union $\nu = \bigsqcup_i \lambda_i \cdot \alpha^{(i)}$.

A part of value 9 in ν requires $\lambda_i \cdot \alpha_j^{(i)} = 9$ for some i, j. Since $\lambda_i \le r = 5$ and $\alpha_j^{(i)} \le k = 4$, the only feasible factorisation is $9 = 3 \cdot 3$, i.e. $\lambda_i = 3$ and $\alpha_j^{(i)} = 3$. Within a single factor $p_{\lambda_i}[h_4]$ with $\lambda_i = 3$: at most one part of $\alpha^{(i)}$ can equal 3 (as $\alpha^{(i)} \vdash 4$; two 3's would give $|\alpha^{(i)}| \ge 6 > 4$). So each factor contributes at most one 9 to ν.

Producing two 9's in ν therefore requires at least two indices i with $\lambda_i = 3$. But $\lambda \vdash 5$ has at most one part equal to 3 (two 3's would give $|\lambda| \ge 6 > 5$). Contradiction. Hence p_ν with $m_9(\nu) \ge 2$ does not appear in $p_\lambda[h_4]$ for any $\lambda \vdash 5$, so $\langle s_\pi[h_4], p_\mu\rangle = 0$ for every μ with $m_9(\mu) \ge 2$. □

Theorem 10 (Theorem B)**.** *For $r = 5$ and $k = 4$: $\Phi_9(t) \mid m_\pi(t)$ for every $\pi \vdash 5$.*

Proof. By Proposition 8, q_9 is a $\mathbb{Z}[\zeta_9]$-linear combination of p_μ's with $m_9(\mu) = 2$. Lemma 9 gives $\langle s_\pi[h_4], p_\mu\rangle = 0$ for each such μ. Hence $m_\pi(\zeta_9) = \langle s_\pi[h_4], q_9\rangle = 0$ for every $\pi \vdash 5$, so $\Phi_9(t) \mid m_\pi(t)$. □

Remark (on the shape of q_e). Small-case computation ($n = 7, e = 5$; $n = 8, 11, e = 3$; and the present $n = 20, e = 9$) is consistent with a uniform statement:

Conjecture (Support conjecture for q_e). Let $n \ge 1$, $e \ge 2$. Then $q_e = \sum_{\lambda \vdash n} \widetilde{f}_\lambda(\zeta_e)\, s_\lambda \in \Lambda_n \otimes \mathbb{Z}[\zeta_e]$ is a $\mathbb{Z}[\zeta_e]$-linear combination of p_μ's with μ of the form $(e^{\lfloor n/e \rfloor},\ \nu)$ where $\nu \vdash (n \bmod e)$.

The Springer-regular cases ($e \mid n$ or $e \mid n - 1$) are trivial: by Corollary 3, q_e is a single monomial of exactly the claimed shape. The content is in the irregular cases. A general proof would presumably follow from a Kraskiewicz–Weyman-style analysis of the fake degree at roots of unity via e-cores and e-quotients; we do not pursue it here. If Conjecture 3.4 holds in general, the argument of Theorem B extends verbatim to any (r, k, e) where the "p_μ has $\lfloor rk/e \rfloor$ parts equal to e" obstruction can be verified from $\lambda_i \cdot \alpha_j^{(i)} = e$ factorisation counts.

4 Verification

All divisibility claims above have been verified by direct SymPy computation implementing Theorem 1. Scripts and logs are archived at `~/projects/probes/2026-08-05-composite-d/` (`scan.py` for the $\Phi_e \mid m_\pi$ scan, `check_qe_support.py` for the ongoing Conjecture 3.4 check at $(20, 9)$). The sanity check (1) passes for every scanned case.

5 Discussion and open questions

What Theorem A proves and does not prove

Theorem A cleanly extends the composite-d warm-up at $k = 2$ that was empirically known but unproven. It does *not* help for $k \geq 3$; the parity argument is genuinely a $k = 2$ phenomenon.

What the correct conjecture should be

Given Theorem C, the correct extension of Theorem 2 to composite d is much more delicate than the "same statement, more divisors" hope of the 2026-07-31 morning WAKE. Empirically at $r = 5$:

$$\text{For which } (k, e) \text{ with } 2 \leq k \leq d-1, e \mid d, e > 1: \quad \Phi_e(t) \mid m_\pi(t) \text{ for all } \pi \vdash r?$$

This appears to depend delicately on both $\gcd(k, e)$ and on Springer-regularity properties of e in S_{rk}; there is no clean single-parameter iff statement of the type known for prime d.

Adjacent open problems

- Complete the empirical scan for $(r, k) = (5, 5), (5, 6), (5, 7), (5, 8)$; identify the pattern of (k, e) pairs for which the strong divisibility holds.
- Verify Conjecture 3.4 at $(n, e) = (20, 9)$ and, ideally, prove it in general.
- Determine whether the "Springer-regular" cases $e \mid n$ or $e \mid n - 1$ always give $[e]_t \mid m_\pi(t)$ for all π, in analogy with the $k = 2$ case.

References

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